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Hybrid automated design of excavation support systems in building projects

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ABSTRACT

Designing excavation support systems (ESSs) typically requires extensive manual data entry from fragmented sources. Previous research has largely focused on recommending a single ESS for an entire project, overlooking site-specific variations. This paper introduces Hybrid Automated Design of ESSs (HADES), a system enabling end-to-end automation from minimal input to a 3D BIM model. By utilizing a site address and excavation boundary, HADES (1) extracts geological and surrounding data from public databases, (2) recommends ESS components and design parameters via a multi-step machine learning pipeline, and (3) automatically generates an early-stage BIM model. Experiments on 313 cases achieved weighted F1-scores of 0.926 for ESS component recommendation and 0.976 for design parameters recommendation. Field applicability was further confirmed through expert-driven reproduction of real-world ESS projects. This demonstrated HADES's potential to streamline the exploration of preliminary designs and enhance early-stage decision-making in construction engineering.

1. Introduction

Excavation support systems (ESSs), comprising retaining walls, water barriers, and lateral supports, are vital for maintaining excavation stability while minimizing urban impacts. However, current ESS design is time-consuming and error-prone due to manual data handling. A method to rapidly generate an ESS building information modeling (BIM) model from minimal early-stage inputs would improve design consistency, reduce errors, and support decision-making. Earlier BIM model availability also enables subsequent tasks such as quantity take-off, 4D simulation, and structural analysis.

Prior studies applied various machine learning algorithms including decision trees [1], self-organizing maps [2], AdaBoost [3], rule induction [4], and case-based reasoning [5] to retaining wall selection, focusing mostly on isolated ESS component. They often rely on numerous factors (sometimes over 30), including inaccessible data (e.g., cost, schedule) or subjective factors (e.g., aesthetics), limiting practical applicability. Furthermore, they overlook component interdependencies and the design parameters (e.g., pile diameters, spacing) required for BIM generation. In addition, these studies do not incorporate actual design practices, including the use of different ESS components depending on the excavation side.

This paper proposes Hybrid Automated Design of Excavation Support Systems (HADES), which enables end-to-end automation from minimal input – excavation boundary and project address – to 3D BIM model generation. HADES is termed "hybrid" as it combines machine learning for ESS component and design parameter recommendation with knowledge-driven parametric modeling for BIM creation.

HADES differs from existing methods in three main aspects (Fig. 1). First, it recommends both ESS components and their design parameters. Second, it automates data collection by retrieving ESS factors, such as surrounding information and geological information, from publicly available sources. Third, it generates early-stage 3D BIM models using these recommendations. Collectively, these aspects enable HADES to streamline the entire process from minimal input to early BIM model generation.

HADES consists of three modules: (1) A public data collector, which retrieves geological and surrounding information from public databases using a site address; it also allows manual input if data is unavailable; (2) An ESS recommender, which suggests retaining walls, water barriers, and lateral supports, along with design parameters; and (3) An ESS BIM model generator, which creates the BIM model based on the previous steps. The proposed method was validated through performance experiments with 1880 real ESS cases and a practicality assessment via the technology acceptance model (TAM) [6].

This paper is organized as follows: Section 2 reviews related work.

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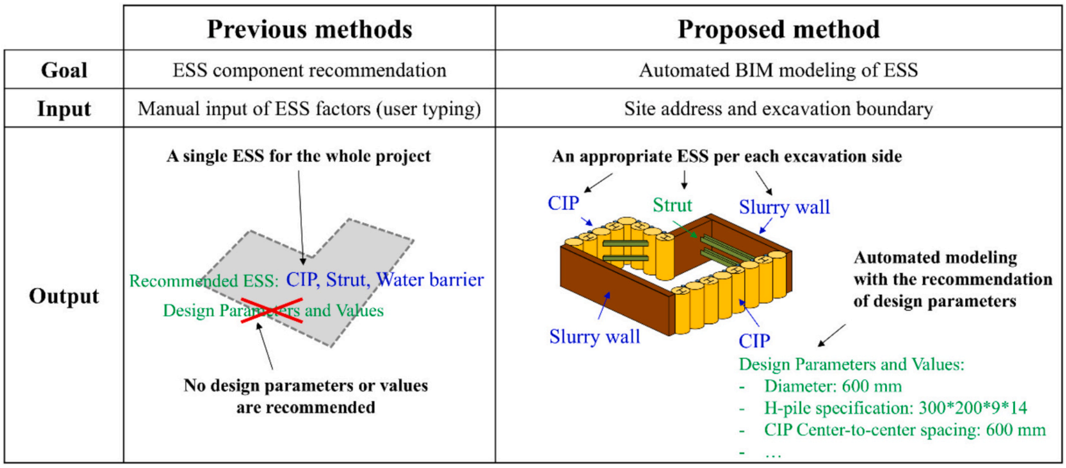

Fig. 1. Comparison of previous methods and the proposed method.

Section 3 introduces the HADES framework. Section 4 describes the experimental setup. Section 5 presents results; Section 6 validates practical applicability. And Sections 7–8 provide the discussion and conclusions.

2. Related work

Several studies have addressed aspects of the HADES framework, yet they remain limited in scope and integration.

First, previous research explored methods to recommend site-appropriate ESS using various algorithms. Case-based reasoning (CBR) was initially developed for retaining wall selection [5], and later improved by incorporating selection rules [7]. With the rise of machine learning (ML) algorithms, including decision trees [1], self-organizing maps [2], AdaBoost [3], prediction accuracy has improved from approximately 70% to over 90%. However, researchers have predominantly focused on retaining walls, with few considering all three components. This line of work also does not fully reflect the ESS design practice, where different ESS components must be combined and their design decisions are interdependent. Furthermore, object-based BIM generation requires associated design parameter values (e.g., pile diameters, spacing) which have not been addressed.

Table 1
Classification of ESS factors by category and source.

Information category	Scource	ESS factor	References
Design information	Drawing or 3D model	Excavation depth	[1–3,8,10]
		Excavation area	[1–3,8–10]
		Length of the excavation side	[9]
		Distance to the opposite excavation side	–
		Number of ground/ underground floors	[1,10]
		Elevation difference	[2,3,9,10]
		Site layout	[3,10]
		Work area	[8,9]
	Experts	Design requirements: stiffness, aesthetic	[9,10]
		Environment: dust, vibration, noise	[8–10]
Surrounding information	Site investigation	Number of adjacent buildings	[1–3]
		Average distance to adjacent buildings	[1–3]
		Ground/underground floors	[1–3]
	Experts	Impact of adjacent buildings	[1]
Geological information	Site investigation	Groundwater: thickness, location	[1–3,8–10]
		Soil layer: type, strength, depth, sandy soil	[1–3,8–10]
		Rock layer: type, thickness, depth	[1–3,8–10]
Others	Construction plan	Cost	[9,10]
		Schedule	[9,10]
		Construction equipment	[9]
	Site investigation	Underground utilities (e.g. gas, electricity, telecommunication)	[8,10]

Second, previous studies required 9 to 31 factors, making manual entry laborious and time-consuming. The factors identified in previous studies could be classified according to the information category (design information, surrounding information, geological information, and others) and according to their source such as drawing or 3D models, experts, site investigations, and construction plans (Table 1). Notably, these factors include both hard-to-obtain factors and subjective factors. While geological data (e.g. rock layer, soil thickness, groundwater) is commonly used in prior studies [1–3,8–10], influential factors such as cost and schedule are often inaccessible due to confidentiality [9,10]. Similarly, environmental impacts often rely on subjective expert judgment, leading to inconsistent results [1,8]. These data limitations not only make factor collection time-consuming but can also render the overall process practically infeasible due to inaccessibility of key factors.

Third, while different ESS component combinations are often applied to each excavation side in practice, previous studies commonly assume a single system for the entire project. In reality, even small sites may require side-specific ESSs due to varying geological or surrounding conditions (Fig. 2).

Fourth, commercial parametric modeling systems, like SmartCon Planner [11] and GEOXD [12], support partially automated ESS BIM generation. However, selecting and modeling components still rely heavily on manual input. Even with compliance to IFC standards, engineers must perform time-consuming data collection and model modification [13].

To address these four limitations, this study improves the ESS design process by enabling end-to-end automation: from automated data collection with minimal input, through comprehensive ESS component and design parameter recommendations, to side-specific BIM model generation. To this end, HADES employs a hybrid framework detailed in the next section.

3. Proposed method

As described earlier, HADES consists of three core modules: the public data collector, the ESS recommender, and the ESS BIM model generator. The overall HADES procedure is shown in Fig. 3. The process begins with the site address and the excavation boundary as initial

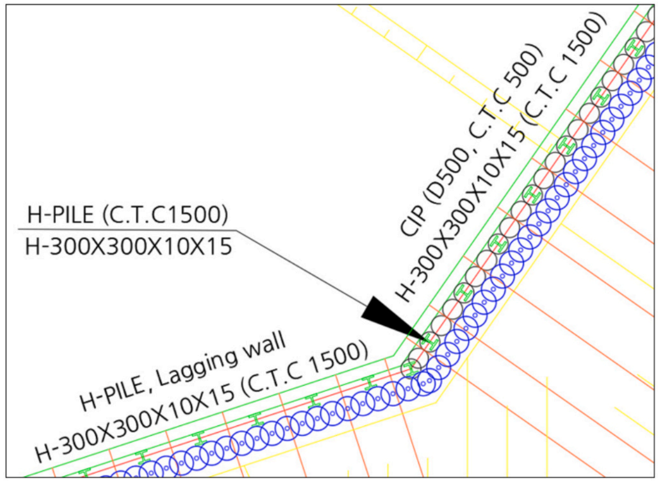

Fig. 2. Example of different ESSs applied to each excavation side under varying geological and surrounding conditions.

inputs. The site address is entered by the user, and the excavation boundary is imported from a 2D drawing in DWG format. The public data collector retrieves ESS factors from public sources and assigns them to each excavation side. The ESS recommender then applies a pre-trained model to recommend ESS components and design parameters for each side. Finally, the ESS BIM model generator produces a BIM model using parametric rules based on the recommended outputs. The following subsections explain each module.

3.1. Public data collector

Currently, public data platforms provide access to infrastructure data, including surrounding and geological information. Notable examples include the U.S. General Services Administration [14], the European data portal [15], and Korea's public data portal [16]. These platforms simplify data acquisition and enable early-stage integration into the design phase. Public data related to ESS design consists of two types: geological and surrounding information, which are retrieved using address-based queries.

Geological information is obtained from site investigation reports using borehole and stratigraphic data. Representative borehole data includes altitude, excavation depth, groundwater level, and geographic coordinates. Stratigraphic data summarizes soil layer composition, including layer names, depths, and thickness. Surrounding information comprises building and site boundaries, reflecting constraints like allowable excavation depth. Building information includes attributes such as floors, area, and usage. By aligning coordinates, buildings can be associated with corresponding site boundaries.

To identify ESS factors, a systematic review was conducted following PRISMA guidelines [17]. Initially, 164 factors were identified from previous studies. Among these, 121 factors were identified as similar or identical to other factors and were consolidated, resulting in 43 factors.

In the screening phase, factors irrelevant to excavation side-based design (e.g., excavation area, site shape) were excluded. While these factors are important for site-based design and overall layout planning, they were omitted to focus on side-specific design decisions, yielding 35 factors for eligibility assessment. In the eligibility assessment phase, factors that were either subjective or not reliably obtainable from public sources were removed. Subjective factors (e.g. environmental metrics, resident safety) were excluded because they depend heavily on expert judgment and are difficult to represent consistently. In addition, project- and contractor-specific factors such as constructability constraints, cost,

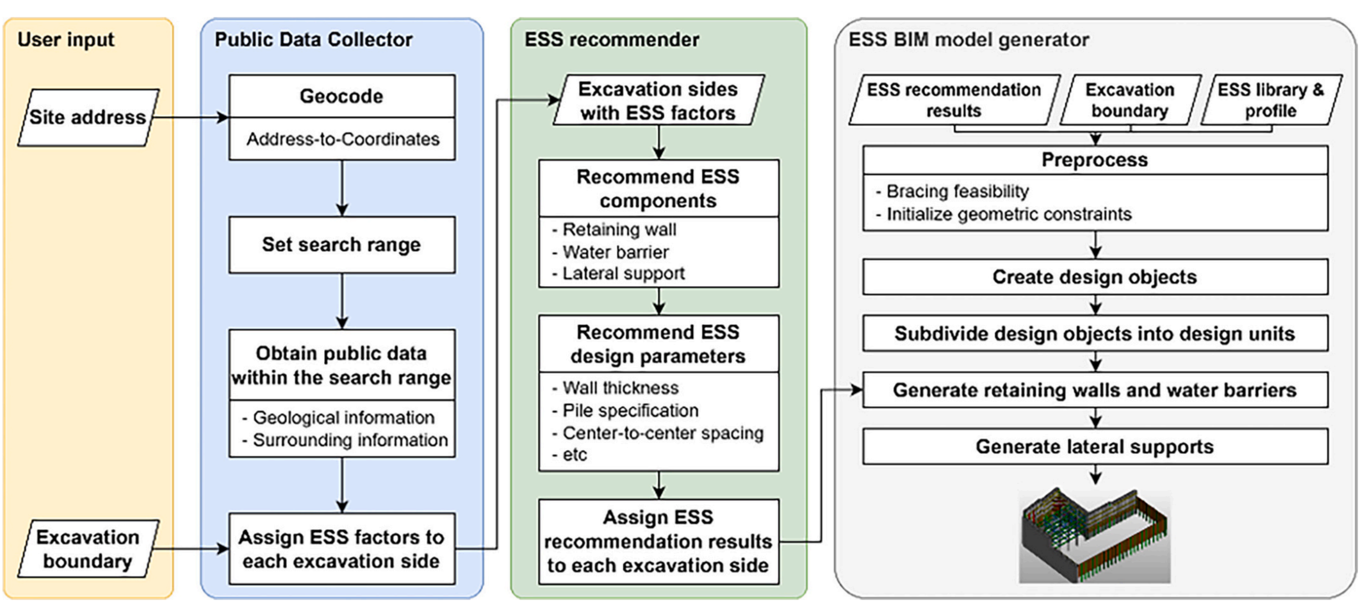

Fig. 3. Procedure of HADES.

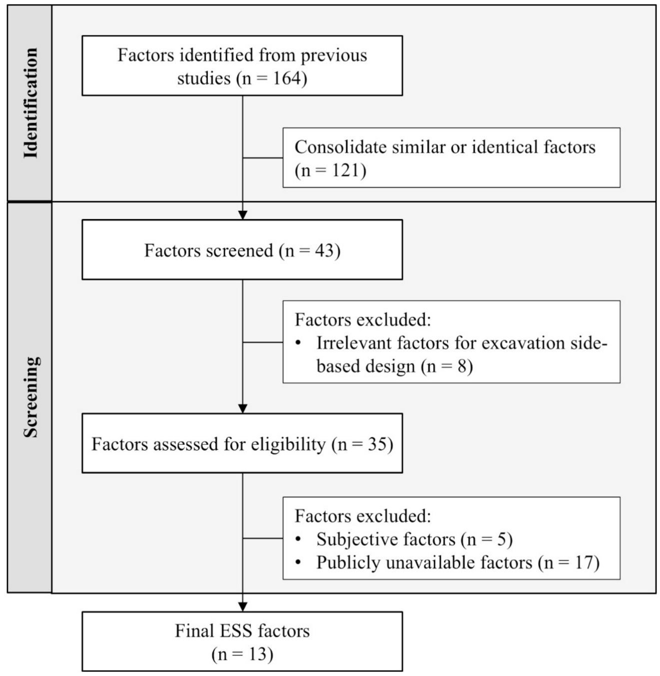

Fig. 4. ESS factor identification and screening process.

Table 2
ESS factors for HADES development.

Category	ESS factors
Design information (D)	D1. Excavation depth
	D2. Distance to the opposite excavation side
	D3. Length of the excavation side
Surrounding information (S)	S1. Number of adjacent buildings
	S2. Average distance to adjacent buildings
Geological information (G)	G1. Groundwater
	G2. Thickness of soil
	G3. Depth to the rock layer (weathered, soft, moderately hard, hard)
	G4. Thickness of the rock layer (weathered, soft, moderately hard,hard)
	G5. Presence of sandy soil
	G6. The bottom rock layer
	G7. Groundwater-bearing layer
	G8. The impact of groundwater

economics, schedule, and material availability were excluded because they are not consistently available in public datasets and often involve confidential know-how (Fig. 4). This systematic screening process resulted in 13 final factors in three categories (Table 2).

The surrounding and geological information were obtained from cadastral and building registry databases. The address is used to retrieve surrounding and geological information, and the excavation boundary is segmented at each bend to assign ESS factors to each excavation side (Fig. 5).

By querying these databases with an address, geographic coordinates are obtained through geocoding [18]. For surrounding information, the "number of adjacent buildings" is determined by generating a neighborhood boundary at a distance twice the excavation depth. Buildings intersected or included by outward normal vectors extended from each excavation side are defined as adjacent buildings. The "distance to adjacent buildings" is the average length of outward normal vectors. For visualization of the surrouding context, building footprints, number of floors, and number of basement levels are also retrieved.

Geological information is collected by searching boring results within a predefined search range of the site, provided as a radius from the site centroid. The location and depth of groundwater, rock layers, and soil layers are extracted. To visualize the stratigraphy, layer depths are interpolated to generate a mesh surface. The excavation side is then positioned on these surfaces, and geological information is assigned based on the midpoint of each excavation side.

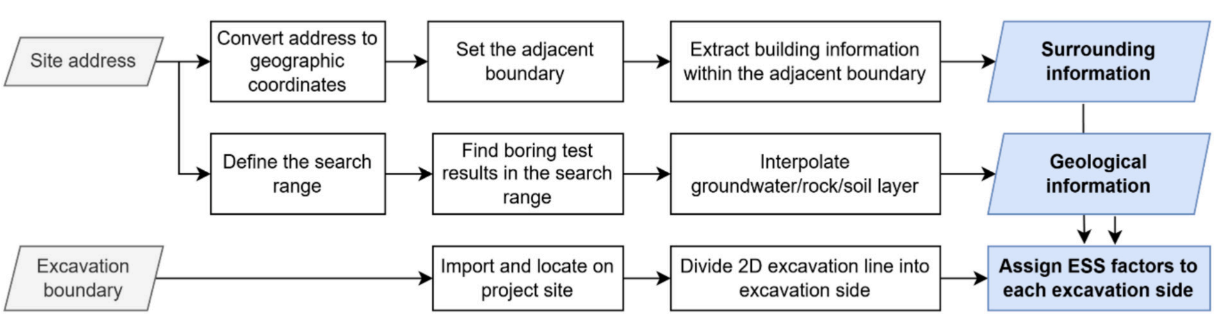

Fig. 5. Public data collection process.

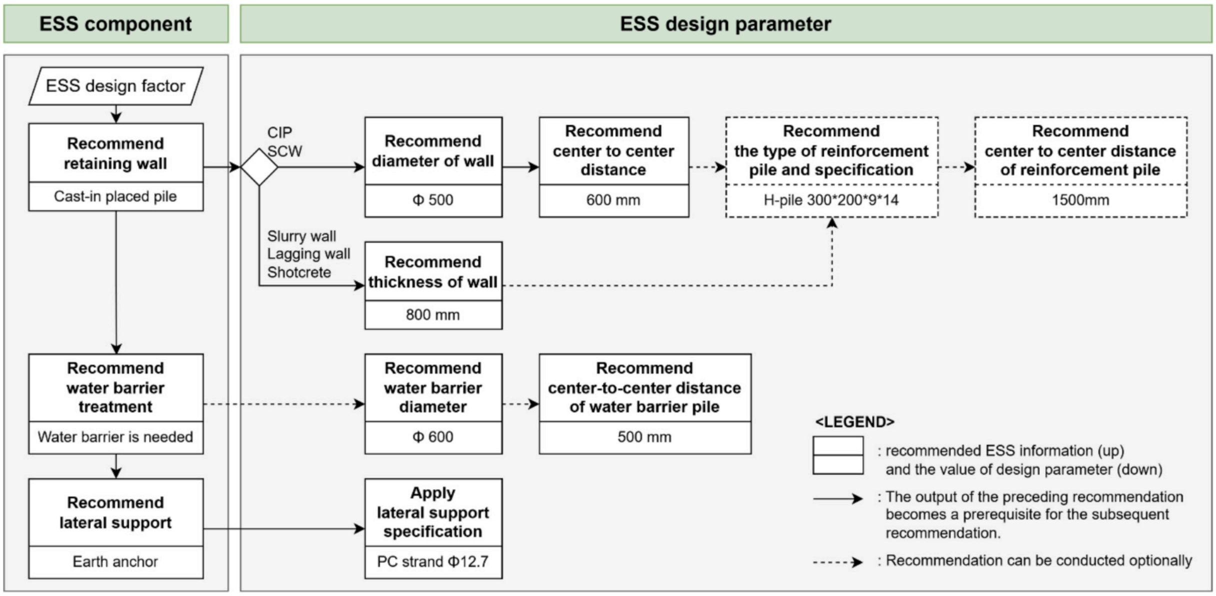

Fig. 6. Multi-step recommendation.

Table 3
ESS components and their types.

ESS component	ESS component types
Retaining wall	Slurry wall, soil cement wall (SCW), cast-in-placed pile (CIP), soldier piles and lagging wall, opencut
Lateral support	Earth anchor, strut, slab, soil nail, rock bolt, raker
Water barrier application (binary)	Required or optional

3.2. ESS recommender

The second module, the ESS recommender, is a pretrained machine-learning model that suggests components and design parameters based on the factors assigned to each side. It follows a multi-step process reflecting practical decision-making, where ESS design involves sequential dependencies: retaining wall selection constrains water barrier applicability, which in turn determines lateral support types and design parameter values. Drawing on this hierarchical nature [19], the approach provides recommendations in two sequential layers: 1) ESS component recommendation and 2) design parameter recommendation. The output of each stage is appended to the input feature set of the subsequent stage. This sequential method has been shown to outperform simultaneous recommendation [20], capturing dependencies while preserving transparency (Fig. 6).

The recommender begins with the retaining wall type. This selection, along with initial factors, informs the water barrier and lateral support recommenders. This ensures context-aware suggestions without relying on fixed rules. After component selection, specific design parameter values are determined. For instance, selecting cast-in-place piles (CIP) leads to recommendations for pile diameter, spacing, and reinforcement, whereas a slurry wall only requires thickness. For water barriers, the model recommends diameter and spacing, while lateral support parameters often utilize standardized project specifications. The component types and design parameters were initially extracted from 2D design documents of multiple ESS design projects, then validiated and refined to meet the ESS design parameters required for BIM model generation. They are summarized in Table 3 and Table 4, respectively.

Table 4
Design parameters for each ESS component type.

ESS component	ESS component type	Design parameter
Retaining wall	Slurry wall	Thickness
	SCW	Pile (diameter, center-to-center spacing), reinforcement beam (type, specification, center-to-center spacing)
	CIP	Pile (diameter, center-to-center spacing), reinforcement beam (type, specification, center-to-center spacing)
	Soldier piles and lagging wall	Thickness, reinforcement beam (type, specification, center-to-center spacing)
Lateral support	Earth anchor	Specification of PC (prestressed concrete) strand
	Strut	Reinforcement beam (type, specification)
	Slab	Thickness
	Soil nail	Diameter
	Rock bolt	Diameter
	Raker	Reinforcement beam (type, specification)
Water barrier	–	Diameter, center-to-center spacing

3.3. ESS BIM model generator

The ESS BIM model generator creates preliminary models using the recommended values and parametric rules based on industry practice. Preprocessing involves two tasks: first, computing "distances to the opposite side" to determine bracing feasibility; and second, initializing site-dependent geological profiles and component libraries. This initializes the geometric constraints required for subsequent modeling.

Design objects are then formalized for every excavation side to prevent the omissions or duplications common in manual configuration. Once valid recommendation outputs are assigned to these objects, the automated placement process is executed. This consists of initializing geographic coordinates, subdividing design objects into design units, and placing ESS components (Algorithm 1). Here, a design object represents an abstract design entity within the boundary box of each excavation side; a design unit is the subdivision of this object into individual units (e.g., walls, water-barrier segments); and an ESS component is the BIM element allocated to each design unit.

Algorithm 1. Preprocessing, initialization, design object creation, and placement.

```
Step 1. Preprocessing & Initialization
        1.1 Compute distances between opposite sides
      W = GetWallListFromSiteProfile()   // W = wallList
          B = ExtractWallBoundaryCurves(W)   // B_w = wallBoundaries
          D_max = CalculateMaxDistancesBetweenWalls(B_w)   // D = distance
          return D_max

        1.2 Initialize site information
      sectionManager = LoadSectionInfoManager()   // section shapes, etc.
          siteInfo = LoadSiteInformation()   // site profile, layers, excavation areas

Step 2. Activate ESS Recommender

Step 3. Design Object Creation
        designObjects = GenerateDesignObjectsForAllSides()
        function GenerateDesignObjectsForAllSides():
          B = GenerateBoundaryCandiatesFromSiteProfile()   // B = candidateBoundary
          for each b ∈ B   // b = boundary
              3.1 Create design object for each excavation side

3.2 Recommend ESS components and design parameters using ESS Recommender
          end for
        end function

        function SetDesignObjectData(d_b, b)   // d_b = designObject
          (1) Find corresponding wall object
          (2) Collect ESS factors per wall
          (3) Run ESS Recommender per wall
          (4) Assign ESS components and design parameter values from ESS Recommender
          (5) Set design types for excavation side:
        end function

Step 4. Placement Stage
        4.1 Initialize geographic coordinate
        4.2 Create design units
        4.3 Place side piles
        4.4 Place retaining walls
        4.5 Place water barrier
        4.6 Place lateral support
```

Detailed placement generates components based on these design units (Algorithm 2). Soldier pile-lagging walls are modeled using lagging thickness and pile sections, while CIP and SCW systems are created by repetitively placing piles at recommended intervals. Slurry walls are modeled as continuous elements. Where required, water barriers are generated behind the retaining wall.

Algorithm 2. Detailed placement for retaining wall and water barrier.

```
Step 5. Retaining Wall Placement
        for each u IN U_d:   // U_d = designUnits, u = unit
          PlaceRetainingWall(u, u.retainingType)
        end for

        function PlaceRetainingWall(u, u.retainingType)
          if type == "SoldierPileLaggingWall": PlaceSoldierPileLaggingWall(u)
          if type == "Shorcrete": PlaceShorcrete(u)
          if type == "CIP": PlaceCIP(u)
          if type == "SCW": PlaceSCW(u)
          if type == "SlurryWall": PlaceSlurryWall(u)
          if type == "OpenCut": PlaceOpenCut(u)

Step 6. Water Barrier Placement
          for each u IN U_d
              if u.waterBarrierType == "Grouting":
                  PlaceGrouting(u)
              end if
          end for
```

Lateral support components are generated after walls and barriers. If wale placement is required, objects are created and offset from the retaining wall boundary using the wall thickness and wale height. Single or double wale arrangements are applied, reinforced by automatically placed brackets. Strut design evaluates geometry and spans to place corner struts, main struts, sub-beams, and post piles. For earth anchors, the system evaluates the side length; if it is at least 500 mm, anchors are computed and placed using wale levels as a reference (Algorithm 3).

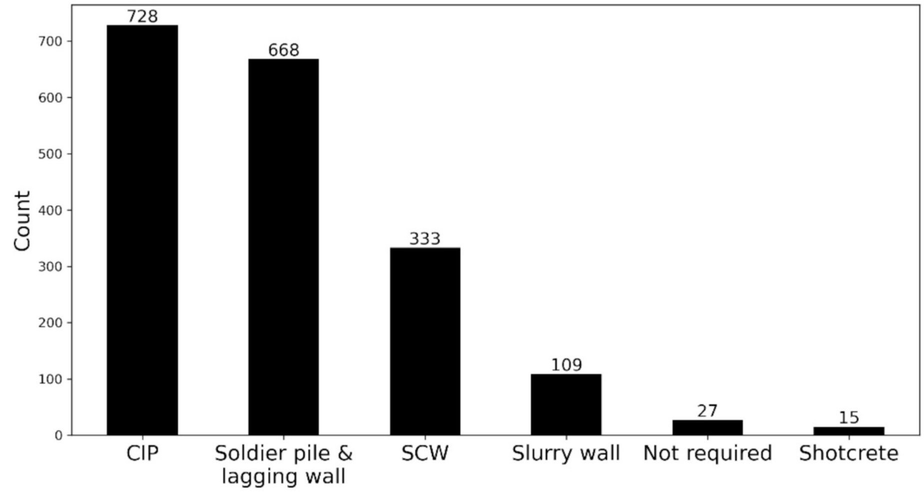

Fig. 7. Distribution of retaining wall cases.

Algorithm 3. Detailed placement of lateral support.

```
Step 7. Wale Placement
        for each u IN U_d:
            if ShouldPlaceWale(u) == FALSE:
                Continue
            end if
            if O_w.t == "Single":    // O_w = waleObject, O_w.t = waleObjectType
                PlaceSingleWale(C_w, O_w)
            else if O_w.t == "Double":
                PlaceDoubleWale(C_w, O_w, O_w.s/2)    // O_w.s = waleObjectSpacing
            end if
            PlaceWaleBrackets(u, O_w)
        end for

Step 8. Strut Design
        8.1 Analyze excavation side and excavation depth
        8.2 Place corner struts (diagonal reinforcement at corners)
        8.3 Place main struts between parallel walls
        8.4 Place sub − beams at strut intersections
        8.5 Place strut strips (main strut members)
        8.6 Place intermediate post piles at strut intersections
        8.7 Bracing placement

Step 9. Earth Anchor Design
            for each u IN U_d:
                if u.wallLength < 500:
                    continue
                end if
                anchorList = DesignAnchorsForUnit()
                for each anchor IN anchorList:
                    ComputeAnchorLevel()
                    ComputeAnchorDirection()
                    AddElementToModel()
                end for
            end for
```

4. Experimental design

This section details the experimental setup by introducing the ESS dataset and the three experiments. The ESS dataset consists of real ESS project data and comprises ESS factors, ESS components, and design parameters. The ESS dataset is used to train the ESS recommender and evaluate its performance. In the address-based ESS factor retrieval experiment, we evaluate the public data collector by assessing address-based ESS factor retrieval, 3D visualization, and ESS factor assignment to each excavation side. In the automated ESS recommendation experiment, we evaluate the ESS recommender by training on the ESS dataset and reporting quantitative performance metrics for component and design parameter recommendation. Finally, in the validation of the generated ESS models, we validate the end-to-end applicability of HADES through a case-study-based comparison between generated ESS BIM models and corresponding as-built cases.

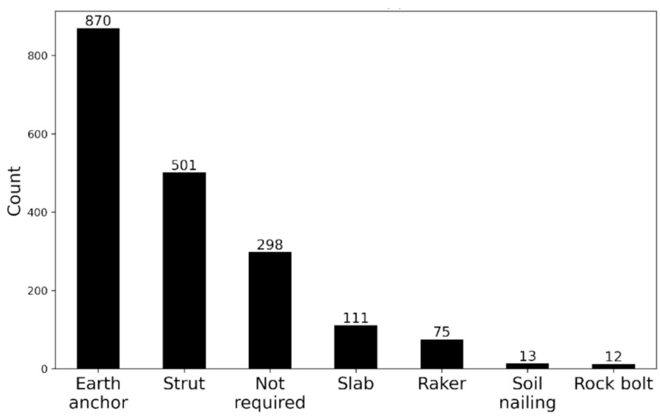

Fig. 8. Distribution of lateral support.

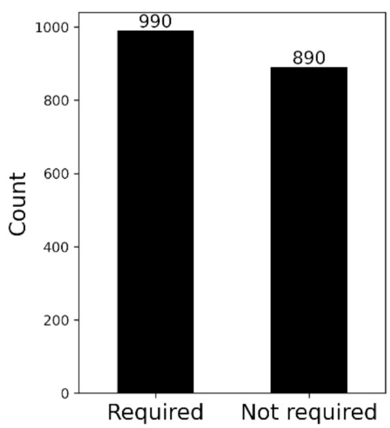

Fig. 9. Distribution of water barrier applications.

4.1. ESS dataset

A total of 1880 excavation side-level cases collected from 124 projects were used to train the ESS recommendation module. Each data entry consists of the components, design parameters, and associated site information assigned to a specific excavation side. Component distribution across the dataset shows that for retaining walls, CIP was most frequent (728 cases), followed by soldier pile–lagging walls and SCW (Fig. 7).

Earth anchors were the dominant lateral support (870 cases), followed by struts (Fig. 8).

Water barrier application is nearly balanced, with 990 "required" cases and 890 "not required" cases (Fig. 9).

4.2. Address-based ESS factor retrieval

To verify the retrieval of geological, building, and design factors, the public data collector underwent 1) a hierarchical test and 2) a qualitative assessment. The hierarchical test assessed geocodability using Korea's lot-number and road-name address systems. This verified if data could be retrieved at each hierarchy level (e.g., metropolitan city to lot number, metropolitan city or road name). Success was measured as a binary (Yes or No) result for factor retrieval.

The qualitative assessment evaluated factor assignment via visualization. Verification of geological information included borehole locations and layer interpolation accuracy. Verification of surrounding information contains the task of checking the location, area, and height of adjacent structures, ensuring alignment with site boundaries. Finally, the assignment of all ESS factors to each excavation side was verified.

4.3. Automated ESS recommendation

The ESS recommender module consists of two sequential stages: ESS component recommendation and design parameter recommendation. To identify the most suitable model for our ESS dataset and to quantify improvements over commonly used baselines, we compared representative algorithms under a consistent evaluation. We included conventional baselines such as Naïve Bayes, Support Vector Machine (SVM), Decision Tree, which have been frequently used in ESS-related prediction tasks. Strong tabular learners based on gradient boosting are also included, such as Gradient Boosting Machine (GBM), LightGBM [21], XGBoost (eXtreme Gradient Boosting) [22], and CatBoost [23], which are widely adopted for structured data. We also tested LLMs including GPT-5 [24] and Gemini 2.5 [25], to benchmark general-purpose foundation models against specialized tabular methods on the same recommendation task. For the LLMs, we provided the same ESS dataset and used instruction prompts to obtain ESS component and design parameter recommendations. The comparison enables both baseline-referenced performance assessment and dataset-specific recommendation of the best-performing recommender.

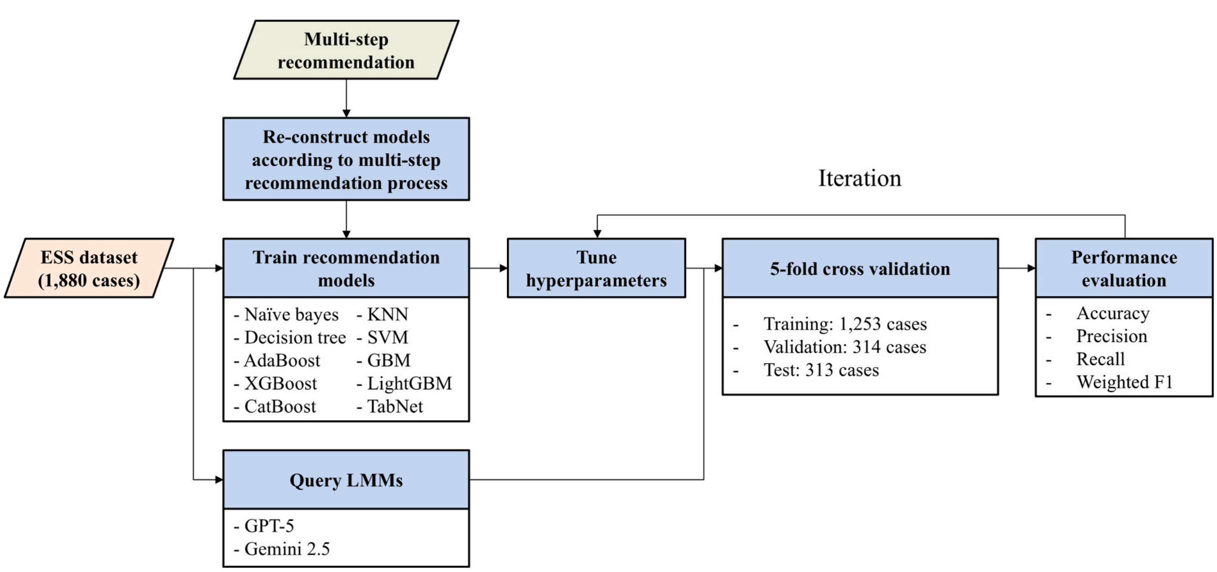

Fig. 10. Model selection workflow for ESS component and parameter recommendation (ML vs LMM).

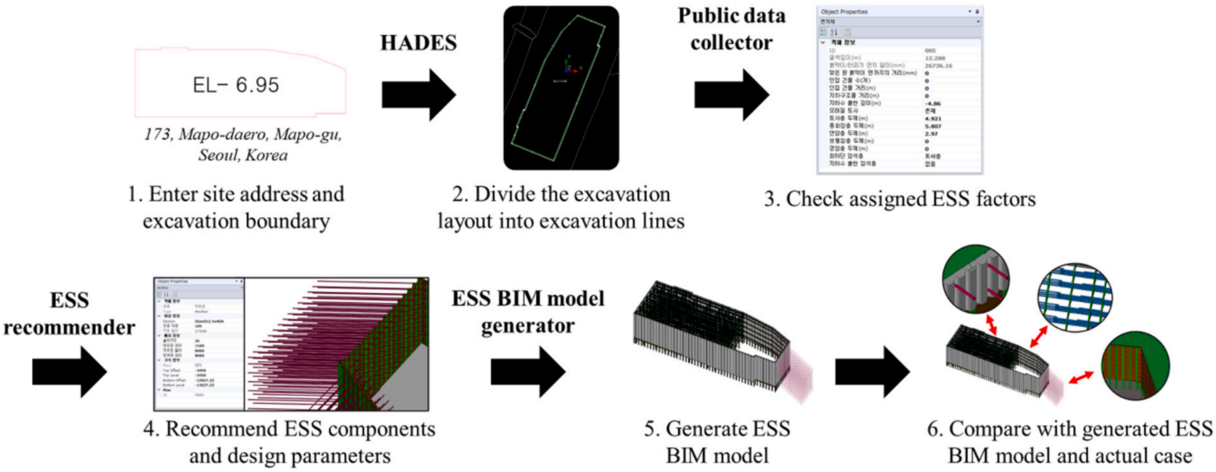

Fig. 11. ESS model generation workflow.

To maximize performance, we tuned key hyperparameters using grid-search library to ensure a transparent and reproducible model selection procedure over a pre-defined search space. The dataset (N = 1880) was first split into an independent hold-out test set (n = 313) and a development set (n = 1567). The test set was kept completely separate from model development and used only for the final, unbiased evaluation. Hyperparameter tuning and model selection were performed exclusively on the development set via five-fold cross-validation. In each fold, the model was trained on four folds and validated on the remaining fold, and the average validation performance across folds was used to select the optimal hyperparameters. After hyperparameter selection, the final model was refit on the full development set and evaluated once on the hold-out test set (Fig. 10).

For the evaluation, four metrics were calculated: Accuracy (Eq. 1), Precision (Eq. 2), Recall (Eq. 3), and Weighted F1 score (Eq. 4). Because the ESS dataset is inherently imbalanced, the weighted F1 score was chosen to account for this by calculating per-class F1 scores and finding their average, weighted by the number of samples. Training was iterated until the recommendation performance was sufficiently improved.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

$$Precision = \frac{TP}{TP + FP} \tag{2}$$

$$Recall = \frac{TP}{TP + FN} \tag{3}$$

$$Weighted\ F1\ score = \sum\nolimits_{i=1}^{N} w_i \bullet F1_i \tag{4}$$

where TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively. w_i is the proportion of samples in class i, and $F1_i$ is the F1-score of class i.

Table 5
Comparison of ESS factor retrieval by address hierarchy in lot-number and road-name address systems.

Address system	Address hierarchy	ESS factors retrieved?
Lot-number address	Metropolitan city / Province	Yes
	City / County / District	Yes
	Town / Township / Neighborhood	Yes
	Block Village	Yes
	Lot number	No
Road name address	Metropolitan city / Province	Yes
	City / County / District	Yes
	Road name	Yes
	Building number	No

4.4. Validation of ESS BIM model generation

We selected four real-world ESS cases for generation, each representing a different scale with varying ESS components and design parameters. To make the generation process explicit, Fig. 11 provides a step-by-step visualization for a representative case, showing 1) the preprocessing outputs from the public data collector, 2) the recommended ESS components and design parameters from the ESS recommender, 3) the resulting BIM model generated by ESS BIM model generator, and followed by a direct comparison with the corresponding as-built case. A demonstration of generating ESS BIM model is available at: https://youtu.be/QrwG-sPLmmM.

Evaluation compared the generated models against the actual cases using the match rate of ESS components (MR_C) and the mean absolute percentage error of design parameters ($MAPE_{DP}$) (Eq. 5–7). The match rate of ESS (MR_C) component indicates the percentage of excavation

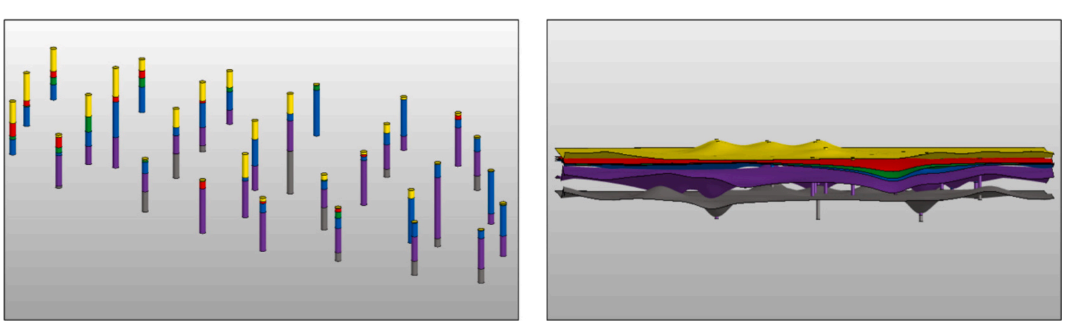

Fig. 12. Identified bore holes (left) and interpolated layers (right).

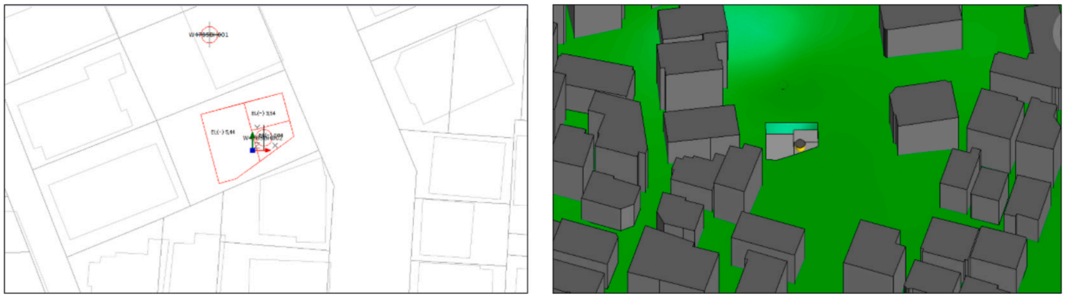

Fig. 13. Identified site boundaries and adjacent buildings in 2D (left) and 3D (right).

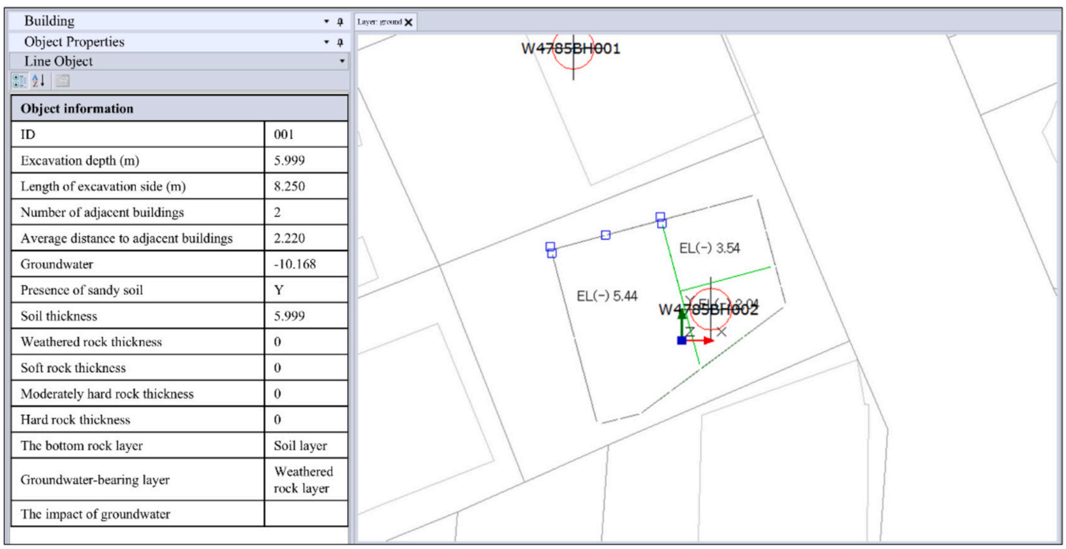

Fig. 14. ESS factors assigned to each excavation side.

Table 6
ESS component recommendation performance: ML vs. LMM.

ESS component	Method	Accuracy	Precision	Recall	Weighted F1	Best model
Retaining wall	LMM	0.2664	0.4255	0.2664	0.2159	Gemini 2.5
	ML	**0.9521**	**0.9525**	**0.9521**	**0.9516**	XGBoost
Lateral support	LMM	0.5015	0.4671	0.5015	0.4208	Gemini 2.5
	ML	**0.8537**	**0.8361**	**0.8537**	**0.8424**	GBM
Water barrier application	LMM	0.6239	0.6252	0.9239	0.6176	GPT-5
	ML	**0.9867**	**0.9868**	**0.9867**	**0.9867**	XGBoost

Table 7
Recommendation performance of design parameters.

ESS parameter	Method	Accuracy	Precision	Recall	Weighted F1	Best model
Retaining wall thickness	LMM	0.0382	0.0262	0.0382	0.0239	Gemini
	ML	**0.9681**	**0.9713**	**0.9681**	**0.9685**	XGBoost
Retaining wall diameter	LMM	0.5601	0.5183	0.5601	0.5328	Gemini
	ML	**0.9761**	**0.9770**	**0.9761**	**0.9753**	CatBoost
Retaining wall center-to-center spacing	LMM	0.4569	0.4446	0.4569	0.4483	Gemini
	ML	**0.9973**	**0.9948**	**0.9973**	**0.9960**	GBM
Water barrier diameter	LMM	0.5292	0.4849	0.5292	0.4872	Gemini
	ML	**0.9894**	**0.9922**	**0.9894**	**0.9906**	GBM
Water barrier center-to-center spacing	LMM	0.5138	0.4300	0.5138	0.4557	Gemini
	ML	**0.9947**	**0.9948**	**0.9947**	**0.9946**	LightGBM
Reinforcement beam type	LMM	0.5707	0.8981	0.5707	0.6686	Gemini
	ML	**0.9920**	**0.9929**	**0.9920**	**0.9922**	LightGBM
Reinforcement beam specification	LMM	0.2994	0.5109	0.2994	0.3470	Gemini
	ML	**0.9681**	**0.9682**	**0.9682**	**0.9677**	LightGBM
Reinforcement beam center-to-center spacing	LMM	0.2265	0.1987	0.2265	0.2083	Gemini
	ML	**0.9309**	**0.9338**	**0.9309**	**0.9299**	GBM

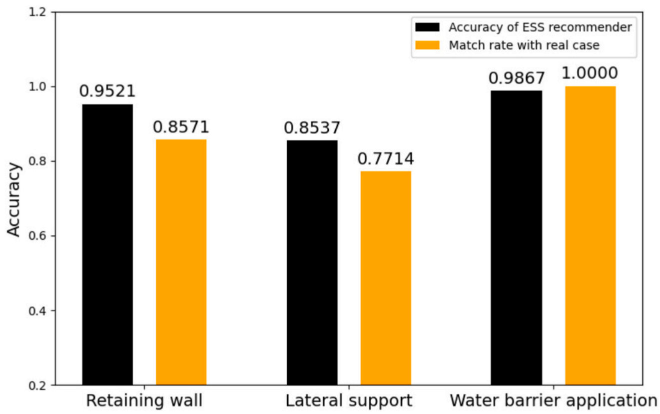

Fig. 15. Accuracy of ESS recommender and match rate with real cases.

sides where the generated ESS component ($\widehat{x}_i$) correctly matched the actual component (x_i). For evaluating $MAPE_{DP}$, the comparison was only performed when the generated and actual ESS components were identical (N_m); cases where the ESS components differed were excluded from the calculation. $MAPE_{DP}$ represents the difference between the generated design parameter value ($\widehat{y}_i$) and the actual design parameter value (y_i), expressed as a ratio. Finally, the error rate (E) between the recommended value ($\widehat{y}_i$) and the actual value (y_i) was evaluated. After that, the suitability for detailed application was assessed by examining the minimum and maximum error values of the design parameters.

$$MR_C = \frac{\sum_{i=1}^{N} I(x_i = \widehat{x}_i)}{N} \tag{5}$$

$$MAPE_{DP} = \frac{100\%}{N_m} \bullet \sum_{i=1}^{N_m} \left| \frac{y_i - \widehat{y}_i}{y_i} \right| \tag{6}$$

$$E = \frac{y_i - \widehat{y}_i}{y_i} \times 100 \quad \text{Eq. (7)}$$

5. Experiment results

This section reports the results of three experiments. First, we present intermediate outputs from the public data collector, including geocoding, 3D visualization, and assignment of ESS factors for each excavation side. We then evaluate the ESS recommender in terms of its ability to recommend ESS components and design parameters, using performance indices. Finally, we assess end-to-end system performance by generating ESS BIM models and comparing them with as-built cases, thereby linking module-level performance to system-level outcomes.

5.1. Information extraction and assignment of ESS factors from public data

The public data collector was evaluated on geocoding coverage, visualization, and factor assignment. Geocoding succeeded across upper hierarchies but failed at the specific lot or building number level because search extents remained unspecified and identifiers were not unique (Table 5).

After coordinate conversion, boreholes within a 200 m radius were rendered in 3D and color-coded by soil and rock layers. Where data was missing, stratigraphic surfaces were generated via interpolation. While this provides an approximation of the subsurface, it enables automated design at the conceptual stage (Fig. 12).

Adjacent buildings and site boundaries were also visualized. In the 3D interface, building outlines were extruded based on floor and height attributes from the public database (Fig. 13). The system successfully assigned these geological and surrounding factors to each segmented excavation side, ensuring that the necessary inputs for recommendation were correctly localized (Fig. 14). These assigned ESS factors constitute the input features used to train and evaluate the ESS recommender in Section 5.2.

5.2. Performance of the ESS recommender

For component recommendation, machine learning (ML) models significantly outperformed LMMs (Table 6). XGBoost achieved the highest weighted F1 scores for retaining walls (0.9516) and water barriers (0.9867). The performance of lateral support was relatively lower (0.8424), which may be attributed to site-specific constraints that are not sufficiently represented in publicly available data, such as buried underground utilities and limited working space. Information on underground utilities is often treated as confidential and is therefore not consistently accessible through public sources. In addition, the selection of lateral support methods is influenced by practical site conditions, including workspace layout, access constraints, and the feasibility of installing earth anchors without encroaching on adjacent properties. Because such factors are generally unavailable in public datasets, the ESS recommender may recommend lateral support methods that differ from those selected by human designers in actual practice.

Design parameter recommendation followed a similar trend, where boosting algorithms (XGBoost, CatBoost, LightGBM) excelled in minimizing loss for structured tabular data. Most parameters achieved F1

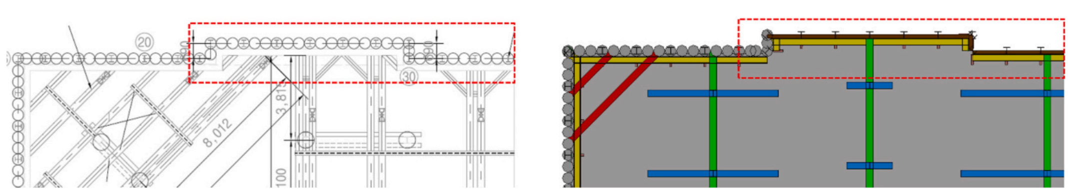

Fig. 16. CIP in real case (left) and soldier pile & lagging wall in reproduced result (right).

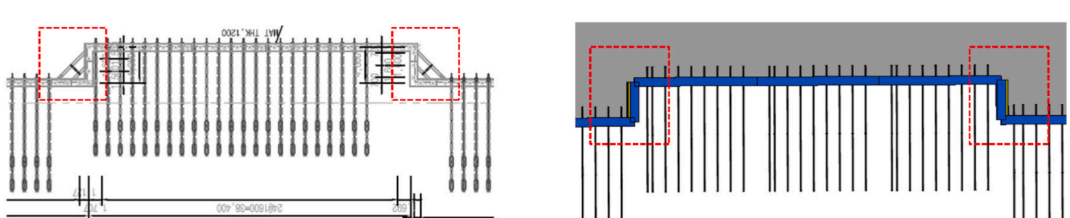

Fig. 17. Strut in real case (left) and no lateral support in reproduced result (right).

Table 8
Evaluation results of recommended design parameters.

ESS component	ESS component type	ESS design parameter	$MAPE_{DP}$	E_{min}	E_{max}
Retaining wall	CIP	Diameter	0%	0%	0%
		Center-to-center spacing	0%	0%	0%
	Slurry wall	Thickness	0%	0%	0%
	Lagging wall	Thickness	0%	0%	0%
	Reinforcement pile	Size	31.95%	0%	109%
		Center-to-center spacing	17.97%	−50%	11%
Water barrier application	Water barrier	Diameter	2.78%	−17%	0%
		Center-to-center spacing	4.17%	0%	25%

scores above 0.9.While slurry wall thickness showed lower accuracy due to class imbalance, the multi-step process successfully captured the interrelationships between components and detailed specifications (Table 7). Among LMMs, Gemini 2.5 was generally superior but still fell far short of the specialized ML models.

Based on the best-performing models, the recommended ESS components and design parameters were used as inputs to the ESS BIM model generator, and the generated ESS BIM model was evaluated against as-built case in Section 5.3.

5.3. Comparative analysis of HADES-generated ESSs against real cases

Using HADES, we generated 35 excavation sides across four real-world cases to measure the component match rate (MR_C) and the design parameter mean absolute percentage error ($MAPE_{DP}$). For component recommendation, the match rates for retaining walls (0.8571) and lateral support (0.7714) were lower than the training accuracies. Water barrier application matched real cases perfectly (Fig. 15).

Discrepancies in retaining wall regeneration occurred when soldier pile and lagging walls were recommended instead of CIP (Fig. 16). While functionally similar, CIP is often used for permanent retention—a factor not currently captured by publicly accessible data. In lateral support, struts were sometimes omitted on short excavation sides where human designers would typically use corner struts for adjacent support (Fig. 17).

Regarding design parameters, CIP dimensions and wall thicknesses showed 0% error. However, reinforcement pile size had a $MAPE_{DP}$ of 31.95%; with some instances, of significant overdesign (up to 109% larger area). The center-to-center spacing of reinf orcement piles was recommended with an overdesigned parameter. It was observed that the E_{min} for the center-to-center spacing of reinforcement pile was −50%, however, this parameter means a distance, this negative error (closer spacing) is understood to be an overdesign. While such overdesign generally enhances structural safety, it may lead to increased material usage and construction costs.

Water barrier parameters remained below 5% $MAPE_{DP}$. However, the E_{min} of the water barrier diameter was −17% and the E_{max} of the water barrier center-to-center spacing was 25%, indicating that under-design had occurred (Table 8). ESS objects with positive errors, which can compromise lateral stability and increase deformation risk. Therefore, spacing-related discrepancies should be carefully reviewed during detailed design to balance safety margins and construction efficiency.

Overall, the end-to-end results indicate that HADES can generate real-world ESS BIM models with high fidelity for several components, particularly water barriers and key retaining wall dimensions (CIP diameter, CIP center-to-center spacing, wall thickness), which show near-perfect matches and low $MAPE_{DP}$. Mismatches of ESS component are mainly observed for retaining walls and lateral supports, likely due to site-specific constraints not captured in public data (e.g., underground utilities, cost, time). Design parameter errors further suggest that reinforcement pile-related and water barrier-related recommendations require careful review, as both conservative overdesign and occasional underdesign can occur. These findings support the use of HADES at the conceptual stage while motivating the integration of additional site-constraint data to improve system-level agreement.

6. Practicality validation

The validation process involved experts executing automated design scenarios in HADES, followed by a qualitative survey and an analysis based on TAM. TAM measures how users accept technology through two primary beliefs: perceived usefulness (PU) and perceived ease of use (PEU). These beliefs determine behavioral intention (BI), which predicts actual usage [26]. Based on this framework, we proposed three hypotheses (Fig. 18):

- H1: PEU positively affects PU.
- H2: PU positively affects BI.
- H3: PEU positively affects BI.

In addition to the TAM questionnaire, experts provided feedback on the specific functionalities provided by HADES.

6.1. Participants

HADES underwent exploratory validation with 17 ESS experts from seven structural engineering firms. The participant group was highly experienced: eleven experts had 10+ years of experience, five had 5–10 years, and one had 3–5 years. The study was approved by an Institutional Review Board (IRB Approval No. 7001988–202,509-HR-2887-02). Due to the sample size, results are interpreted as exploratory evidence.

6.2. Measurements and reliability assessment

Following a demonstration and tutorial, 17 experts performed an ESS

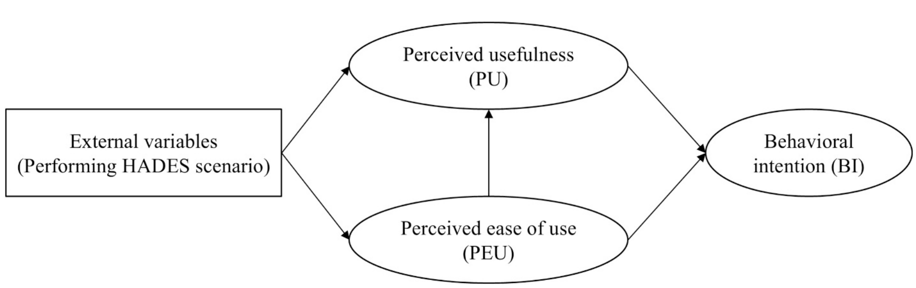

Fig. 18. TAM framework.

Table 9
Questionnaire for ESS validation.

Category		Code	Contents
Technology acceptance model (TAM)	Perceived usefulness (PU)	PU1	I think I could finish my work faster by using this system.
		PU2	I believe this system would enhance my performance at work.
		PU3	I would be more efficient in my job by using this system.
		PU4	Using this system would increase my productivity.
		PU5	I believe this system would make it easier to handle my tasks.
		PU6	I would find this system useful in my job.
	Perceived ease of use (PEU)	PEU1	I think it would be easy for me to learn this system.
		PEU2	I believe I could easily make this system perform the tasks I want.
		PEU3	My interaction with this system is clear and understandable.
		PEU4	I think that this system would be straightforward and easy to comprehend.
		PEU5	I feel it would be easy for me to become proficient with this system.
		PEU6	I think that this system would be easy to work with.
	Behavioral intention (BI)	BI1	I would consider using this system in the future.
		BI2	I intend to use this system going forward.
		BI3	I will continue to use this system.
		BI4	I will recommend this system to others.
Functionality		F1	Users can load geological information through address input.
		F2	The loaded geological information can be used to visualize a stratum model.
		F3	Users can retrieve and visualize information on surrounding buildings by entering an address.
		F4	The relevant data (geological information, building information) for the automated design has been properly allocated to each side of the excavation boundary.
		F5	For each excavation side, the ESS—including the retaining wall, water barrier, and lateral support—have been appropriately applied.
		F6	The automatically generated draft model was completely formed, including all objects such as the retaining wall, earth anchors, and auxiliary materials.
		F7	The design parameters (e.g., thickness, diameter, spacing, specifications) for the auto-generated objects have been appropriately set.

Table 10
Descriptive results of TAM.

Category	Mean	SD	95% CI_Low	95% CI_High
Perceived usefulness	4.059	1.059	3.514	4.603
Perceived ease of use	4.108	0.870	3.661	4.555
Behavior intention	4.162	0.931	3.683	4.640

Table 11
Reliability analysis results for the TAM questionnaire.

Category	The number of questions	Cronbach's α	Composite realiability (CR)
Perceived usefulness (PU)	6	0.989	0.991
Perceived ease of use (PEU)	6	0.955	0.973
Behavior intention (BI)	4	0.982	0.988

Table 12
Average variance extracted (AVE) value of TAM constructs.

Category	AVE (>0.50)
Perceived usefulness (PU)	0.951
Perceived ease of use (PEU)	0.858
Behavior intention (BI)	0.954

Table 13
Heterotrait-Monotrait Ratio.

Construct combination	HTMT (< 0.90)
PU-PEU	0.842
PU-BI	0.946
PEU-BI	0.853

BIM model generation scenario using HADES. They then completed a 23-item TAM questionnaire and functionality survey (Table 9) using a 5-point Likert scale. Statistical analyses, including reliability tests and structural equation modeling, were performed using SPSS and AMOS.

TAM descriptive results from 17 ESS experts indicate generally favorable attitudes toward HADES on a 5-point Likert scale (1 = "Strongly disagree" and 5 = "Strongly agree"). Construct means were PU = 4.059 (95% confidence interval: 3.514–4.603), PEU = 4.108 (95% confidence interval: 3.661–4.555), and BI = 4.162 (95% confidence interval: 3.683–4.640), with slightly higher central tendency for behavioral intention relative to usefulness and ease of use (Table 10). These descriptive findings suggest a positive willingness to apply HADES in practice, while interpretations remain exploratory given the small expert sample.

Prior to hypothesis testing, reliability and convergent validity were assessed. Reliability refers to the degree to which a measurement instrument is stable and consistent, indicating whether if produces similar results when repeatedly applied under the same conditions. To verify whether the questionnaire items used in this study were statistically reliable, Cronbach's α coefficients were examined. Cronbach's α is an indicator used to measure the internal consistency of the items within a construct [27]. The reliability of the measurement was assessed by determinig whether the items used to measure PU, PEU, and BI were highly correlated and conceptually consistent with each other.

A Cronbach's α value above 0.6 is acceptable for exploratory research, above 0.8 for basic academic research, and above 0.9 for studies involving critical decision-making [28]. The results of this validation showed that the Cronbach's α values for PU, PEU, and BI among expert respondents all exceeded 0.95. Additionally, the composite reliability (CR) values for all constructs were also greater than 0.95, indicating that the measurement items demonstrated excellent internal consistency (Table 11). Average variance extracted (AVE) values were above 0.85 for PU, PEU, and BI, confirming strong convergent validity (Table 12). Although the heterotrait-monotrait ratio (HTMT) ratio between PU and BI slightly exceeded the 0.9 threshold (Table 13), it was considered acceptable given the strong theoretical association between the two constructs in TAM-based models.

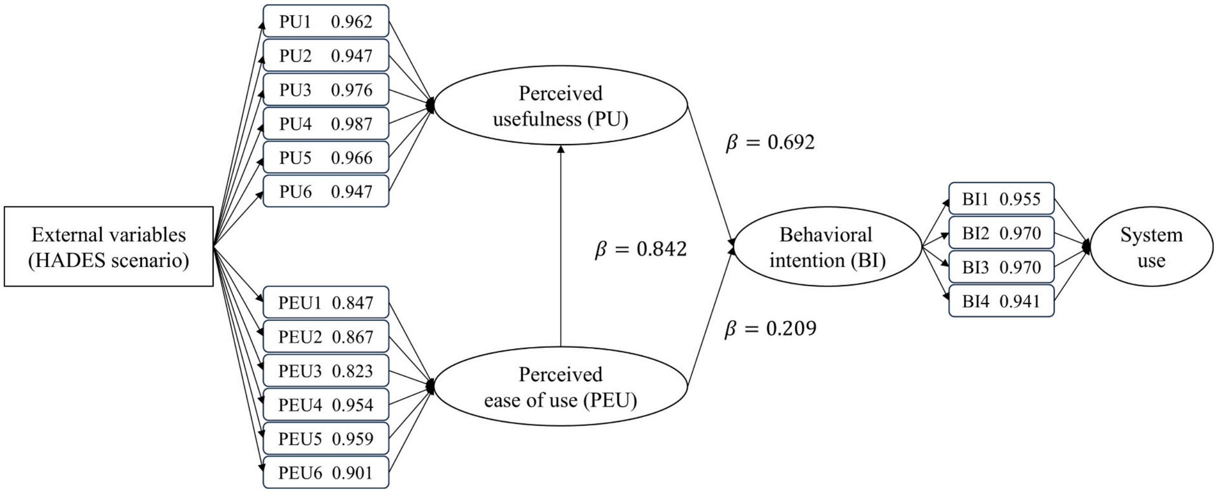

Fig. 19. Technology acceptance model (TAM) applied to the HADES scenario.

Table 14
Experts' opinions of HADES functionality.

ESS module	Feedback	Comments
Public data collector	Positive	- Visualization of geological and building information was sufficiently detailed. - Visualize groundwater levels enabled more informed decision-making for water barrier recommendation. - Visualization of adjacent building can support construction planning tasks (e.g. securing workspace).
	Complementary	- The accuracy and reliability of the interpolation algorithm should be ensured. - Underground utility information (electricity, water & sewage, etc.) can be helpful in decision making. - The public data itself may be incorrect or not up to date, so double-checking is necessary.
ESS recommender	Positive	- Recommendation of ESS alternatives while allowing users to make the final selection is a reasonable approach. - The idea of applying different ESS component and design parameters depending on the surrounding environment of each excvation side is acceptable.
	Complementary	- The increase of training data can enhance the reliability of HADES. - By combining with structural analysis or calculation-based recommendation, it would be more suitable in practice. - If the preference for ESS could be reflected by each institutional and country, it would be even more useful.
ESS BIM model generator	Positive	- Users can directly modify generated ESS models and adjust design parameters. - It is useful to generate an initial ESS model by using only drawings and a site address.
	Complementary	- Clarifying the rationale behind the recommendation of specific ESS can convince potential users. - ESS elements (e.g. bracing, anchors) were sometimes placed with unrealistic spacing. - More advanced parametric modeling algorithm could shorten the design period.

6.3. Hypothesis testing

Regression-based structural analysis confirmed that PU has a strong, direct effect on BI ($\beta = 0.692$, $p < 0.001$). While PEU significantly influenced PU ($\beta = 0.842$, $p < 0.001$), it did not have a significant direct effect on BI. The analysis result indicates that H1 and H2 are supported, whereas H3 is rejected. Furthermore, user adoption is primarily driven by functional benefits, with ease of use acting as a secondary factor that enhances perceived value (Fig. 19). The model explained 90.6% of the variance in behavioral intention.

Table 15
Comparison between prior limitations and this study.

	Prior limitations	Our approach
1	Focusing on ESS component prediction	Recommendation of ESS components and their design parameter values considering their interrelationship
2	Single ESS for entire project	Excavation side-specific ESS recommendation
3	Parameteric modeling of ESS based on predefined ESS component by designer	End-to-end automated ESS model generation
4	Manual input of ESS factors	Public data-based automated input of ESS factors

6.4. Expert review of system functionality

Following the execution of the ESS design scenario, the researchers collected comprehensive expert feedback regarding the Public Data Collector, ESS Recommender, and ESS BIM Model Generator modules within HADES. The Public Data Collector was highly rated for its detailed visualization of geological and surrounding information, which was deemed sufficient for early-stage design requirements. Specifically, the visualization of groundwater levels facilitated more informed decisions regarding water barrier selection, while the spatial representation of adjacent structures showed promise for supporting logistical planning and workspace management. However, the experts pointed out that the system's reliance on interpolation in data-sparse areas necessitates a highly reliable algorithm to maintain confidence. They also recommended the inclusion of underground utility data, such as electrical conduits and drainage networks, to provide a more holistic view of the site conditions.

Regarding the ESS Recommender, experts appreciated its ability to propose localized alternatives while maintaining the user's role in the final selection process. While the model's side-specific recommendations were viewed as acceptable, the lower accuracy of lateral support suggestions—likely due to dataset limitations—remains a point for improvement. To increase practical utility, the experts suggested that combining these data-driven insights with formal structural analysis or regional design rules would make the system more robust for real-world

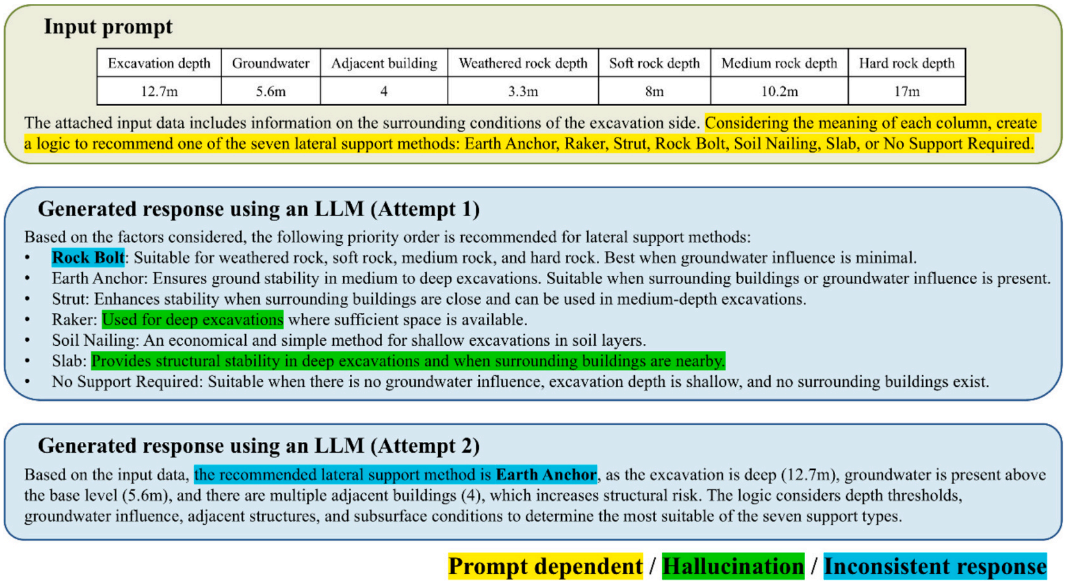

Fig. 20. Example output of LMM for lateral support recommendation.

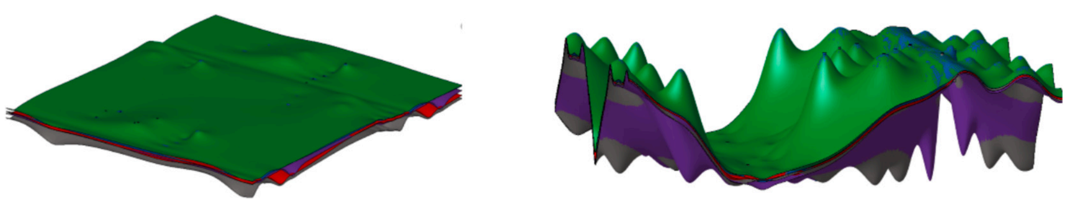

Fig. 21. Strata generated using geological information.
(left: correct information, right: incorrect information).

application.

The ESS BIM Model Generator was positively received for its capacity to significantly reduce design time by allowing for direct manual adjustments to the automatically generated 3D models. Its ability to produce a draft model from minimal inputs, such as a site address, was recognized as a significant asset for the conceptual phase of a project. To enhance user confidence, the experts suggested implementing "explainable" features that clarify the reasoning behind specific AI-generated design choices. Additionally, they identified a need for more sophisticated parametric algorithms to address occasional issues with unrealistic spacing in bracing and anchors, ensuring that the generated models align more closely with engineering standards. These collective insights into the system's strengths and required improvements are summarized in Table 14.

7. Discussion

This paper demonstrates both the technical feasibility and practical effectiveness of automating preliminary ESS design through the integration of public data, machine learning-based recommendation models, and BIM-based parametric modeling. The HADES successfully generated ESS models from minimal input (site address and excavation boundary), achieving weighted F1-scores of 0.951 for retaining walls, 0.842 for lateral supports, and 0.986 for water barrier applicability. Parameter prediction achieved weighted F1-scores ranging from 0.929 to 0.996, demonstrating that the system can reproduce expert-level design judgment. These results indicate that learning-based approaches can achieve design consistency comparable to expert judgment even in complex multi-stage design problems.

ESS design inherently involves sequential dependencies: the retaining wall type constrains lateral support type, which subsequently affects water barrier applicability and the corresponding design parameters in a hierarchical decision structure. As summarized in Table 15, previous studies largely treated the automated ESS design as a single component prediction problem and did not cover interrelationships among ESS components or provide the design parameter values required for ESS BIM generation. In contrast, this study considered interrelationships through a multi-step recommendation and then translated the recommendation results into BIM objects via parametric generation. Whereas prior studies commonly assumed a single ESS for an entire project, HADES produces excavation-side-specific recommendations to reflect surrounding conditions. Moreover, unlike conventional parametric modeling workflows that require designers to preselect components and manually configure modeling inputs, HADES automates BIM element placement and parametrization from the ESS recommender outputs. Finally, ESS factor input is simplified by replacing manual entry with automated retrieval based on the side address and excavation boundary drawing.

However, several considerations merit careful discussion regarding practical deployment and future research directions. First, as clarified in the PRISMA-based factor screening (Section 3.1), this study considers only ESS factors that are consistently accessible from publicly available sources. Accordingly, project- and contractor-specific factors such as constructability constraints, cost, and schedule, although influential in practice, were not included as inputs to HADES and may lead to differences between HADES recommendations and actual design outcomes. For example, while HADES may recommend different ESS components for each excavation side optimized for local ground conditions, practitioners often standardize to a single method to improve equipment mobilization efficiency and simplify construction coordination. This discrepancy indicates that HADES performs technical optimization under public data constraints, whereas human designers conduct multifaceted optimization under broader project contexts. In interviews, practitioners perceived HADES not as a final design generator but as a "design assistance tool that rapidly suggests alternatives not previously considered." which aligns with the system's human-in-the-loop design philosophy. That is, HADES implements a human-AI collaboration model that explores the design space and presents diverse technical alternatives to support designers' creative thinking, while leaving final decisions to human experts who can comprehensively judge project-specific constraints.

Second, although the proposed recommendation models perform well, they remain black boxes to users. Feature importance scores offer limited insight, which hinder trust. Case-specific explainability is therefore essential in AI-assisted design. Techniques such as SHAP (Shapley additive explanations) provide instance-level reasoning, allowing engineers to understand the contribution of each ESS factor to a specific recommendation [29,30]. This paper focused on demonstrating the technical feasibility of ESS design automation, and adding an explainability layer on this foundation will be the next research stage. Large language models (LMMs) such as ChatGPT, LLaMA, and Grok show premise for generating design rationales [31,32]. Despite their potential, LMMs face challenges in engineering contexts, including flawed design logic, lack of project-specific reasoning, hallucinations, and prompt dependency [33,34]. Even when syntactically correct, LMM outputs may lack the depth and clarity required for expert-level ESS decision-making (Fig. 20). Future research will develop a hybrid explanation system satisfying both statistical accuracy and engineering logic by combining SHAP-based quantitative analysis with retrieval-augmented generation utilizing validated engineering knowledge bases. This approach aims to deliver case-specific, human-readable explanations about recommendations to engineering rationale, thereby enhancing trust and practical adoption.

Third, public data quality represents a fundamental constraint on system performance. In some cases, outdated public data misaligned with site conditions and caused inappropriate visualization, which resulted in inappropriate recommendations (Fig. 21). Public datasets in Korea are generally updated on 1 to 5-year cycles and may not immediately reflect recent developments or localized changes. However, this is not a problem unique to HADES but rather a constraint inherent to public data-based automation systems in general. What matters is that the 0.80 to 0.98 range of performance achieved in this study presents the upper bound of automation achievable at current public data infrastructure levels. This implies scalability potential where system performance can improve as data quality improves in the future. Moreover, from a practical workflow perspective, the automatically generated drafts provided by HADES offer a structured starting point for cross-validation with site data, making error detection and correction more efficient than traditional fully manual approaches.

8. Conclusions

This paper addressed the limitations of existing ESS research, which largely focused on ESS component prediction, mostly retaining walls, and required extensive, subjective input variables. Furthermore, previous studies overlooked both the interdependencies among ESS components and the design parameters necessary for 3D model generation. To fill this gap, we proposed HADES, a hybrid method that enables end-to-end automated design from minimal input to early stage ESS design and BIM model generation.

HADES achieved weighted F1-scores of 0.926 for component recommendations and 0.976 for design parameter suggestions, HADES demonstrates expert-level design capability. Validation using real-world ESS cases and a TAM-based usability assessment confirmed that the HADES is both technically reliable and practically deployable. This work has four key contributions:

First, HADES enables *end-to-end automation* from minimal input to a 3D BIM model while previous studies have focused only on ESS recommendation.

Second, unlike previous studies that were limited to recommending a single component ESS for the entire project based on manually curated datasets, this work proposes a multi-step machine learning pipeline that can recommend ESSs tailored to each excavation side. This is particularly crucial for large excavation projects, which typically deploy varying ESSs depending on the conditions of each excavation side.

Third, traditional and previous methods require manual collection and input of sometimes over 30 pieces of information to design ESSs from fragmented sources. Moreover, many of these inputs were difficult to acquire due to the limited accessibility (e.g., cost, schedule) or suffered from subjectivity (e.g., aesthetic, noise). This study creates a fully functional ESS BIM model using the minimum number of inputs, most of which can be collected from public data sources.

Fourth, by generating BIM draft models from recommended information, this research demonstrates how data-driven design can be systematically implemented in early-stage design processes while incorporating real-world engineering workflows.

From an industry perspective, HADES enables engineers to generate preliminary ESS draft models with minimal input, only an excavation boundary and site address. Validation using real-world ESS projects confirms the framework's effectiveness in bridging AI, public data infrastructure, and BIM-based modeling, addressing both academic and practical gaps while providing scalable foundation for advancing intelligent design processes in ESS.

Despite its contributions, several limitations remain. First, no engineering validation method has been incorporated yet. Second, although we believe that the proposed method has generalizability to support automated ESS design in other regions, the validation in this study was conducted only using the datasets produced in South Korea. Third, although HADES demonstrates high accuracy, this does not imply design completeness, as ESS design requires the simultaneous consideration of selection factors beyond technical feasibility. Therefore, accurate recommendations alone cannot fully represent the decision-making process inherent in practical ESS design.

Building upon these initial design drafts, subsequent developments could integrate advanced functionalities including quantity take-offs, cost estimation, construction scheduling, and multi-objective optimization considering technical, economic, and environmental factors simultaneously. By demonstrating the feasibility of end-to-end automation from public data extraction to parametric BIM generation, this research establishes a framework that can accelerate digital transformation across construction engineering domains. As public data infrastructure continues to improve and AI interpretability techniques mature, systems like HADES will become a key player in knowledge-based design practices, fundamentally transforming how engineers approach preliminary design in the construction industry.

CRediT authorship contribution statement

Hyunsung Roh: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **Ghang Lee:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Miyoung Uhm:** Writing – review & editing, Methodology. **Suhyung Jang:** Writing – review & editing, Investigation. **Yonghan Kim:** Writing – review & editing, Software. **Sungil Ham:** Writing – review & editing, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Some or all data, models, or code that support the findings of this study are available from the corresponding author upon reasonable request.

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